Surface Pattern Recognition Using Shape Subspace

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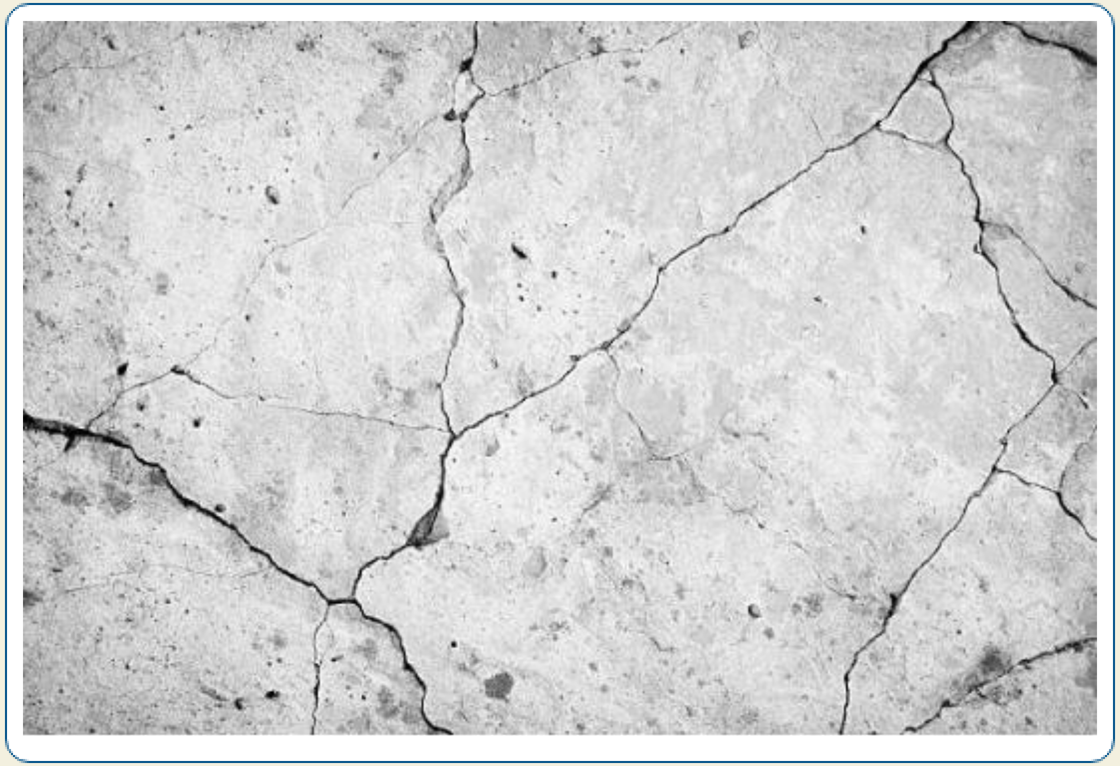
Contents

Research Problem

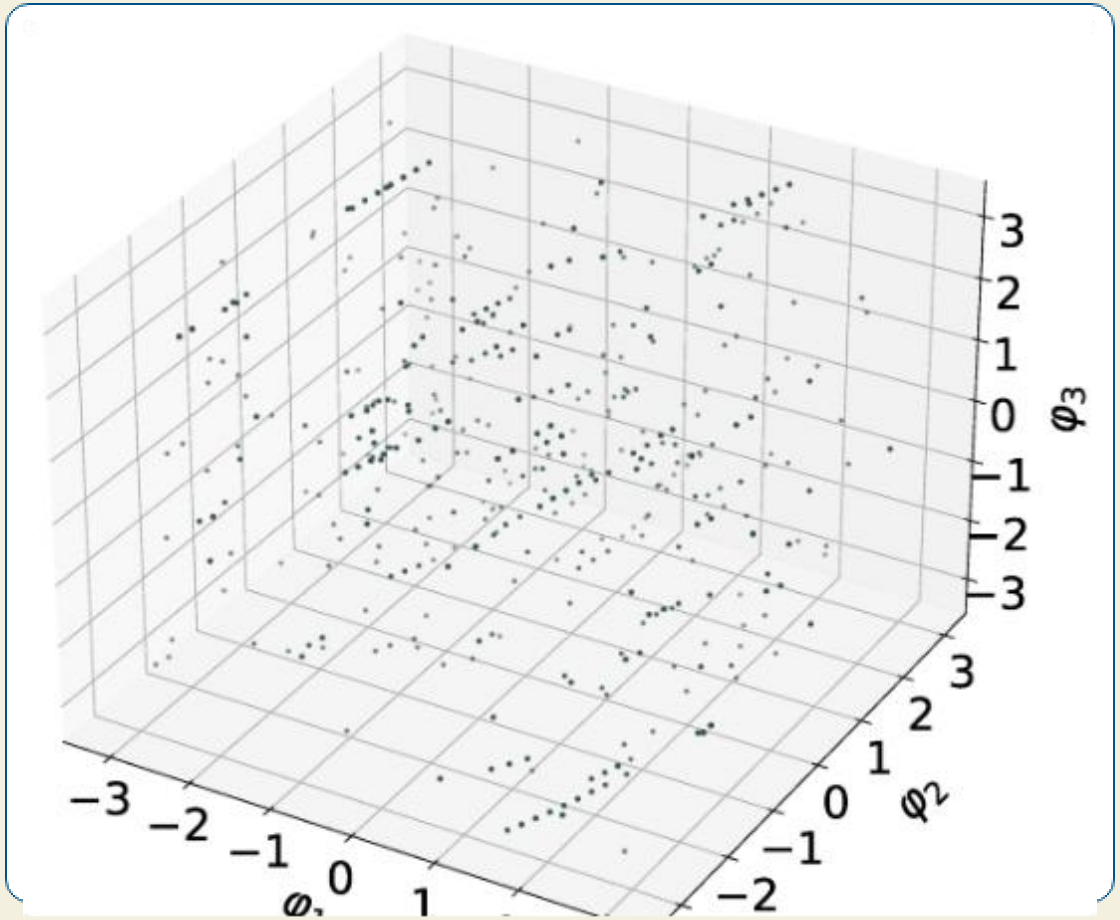
How can Subspace-based methods be used to detect surface irregularities in Point Cloud data?



Background: Point Cloud

Point Cloud processing involves handling 3D data represented as a collection of points in space (x, y, z) coordinates.

Data can include coordinates, color (RGB), and normal vectors.



Background: Shape Subspace

- A mathematical representation of 3D structures based on the relative positions of feature points.
- Used for tasks such as object recognition, motion analysis, and biometric authentication.
- Provides an efficient way of representing high-dimensional data.

$$\frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial f_1(x)}{\partial x_1} & \frac{\partial f_1(x)}{\partial x_2} & \frac{\partial f_1(x)}{\partial x_3} & \frac{\partial f_1(x)}{\partial x_4} & \cdots & \frac{\partial f_1(x)}{\partial x_n} \\ \frac{\partial f_2(x)}{\partial x_1} & \frac{\partial f_2(x)}{\partial x_2} & \frac{\partial f_2(x)}{\partial x_3} & \frac{\partial f_2(x)}{\partial x_4} & \cdots & \frac{\partial f_2(x)}{\partial x_n} \\ \frac{\partial f_3(x)}{\partial x_1} & \frac{\partial f_3(x)}{\partial x_2} & \frac{\partial f_3(x)}{\partial x_3} & \frac{\partial f_3(x)}{\partial x_4} & \cdots & \frac{\partial f_3(x)}{\partial x_n} \\ \vdots & & & & & \\ \frac{\partial f_m(x)}{\partial x_1} & \frac{\partial f_m(x)}{\partial x_2} & \frac{\partial f_m(x)}{\partial x_3} & \frac{\partial f_m(x)}{\partial x_4} & \cdots & \frac{\partial f_m(x)}{\partial x_n} \end{bmatrix}$$

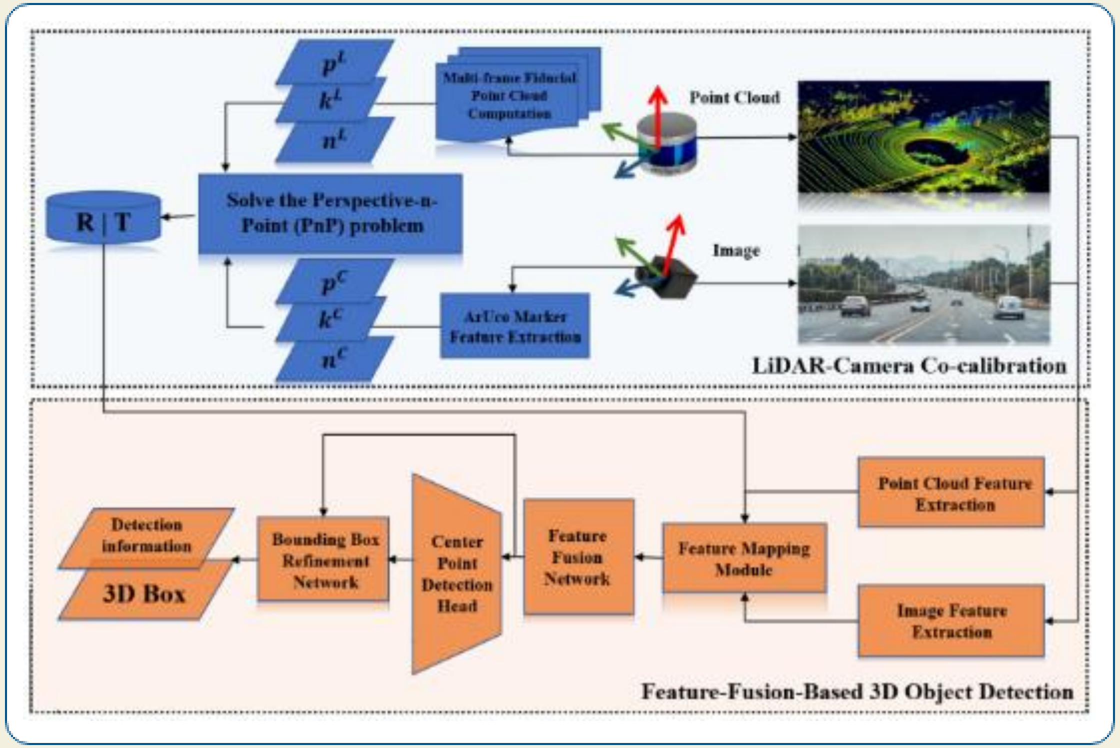
$$\frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial sin(24x_1x_2x_3)}{\partial x_1x_2x_3} & \frac{\partial sin(24x_1x_2x_3)}{\partial x_1x_2} & \frac{\partial sin(24x_1x_2x_3)}{\partial x_3} \end{bmatrix}$$

$$\frac{\partial f}{\partial x} = \begin{bmatrix} 24cos(24x_1x_2x_3) & 24x_3cos(24x_1x_2x_3) & 24x_2x_3cos(24x_1x_2x_3) \end{bmatrix}$$

Related Research: Factorization

How is Shape Subspace computed?

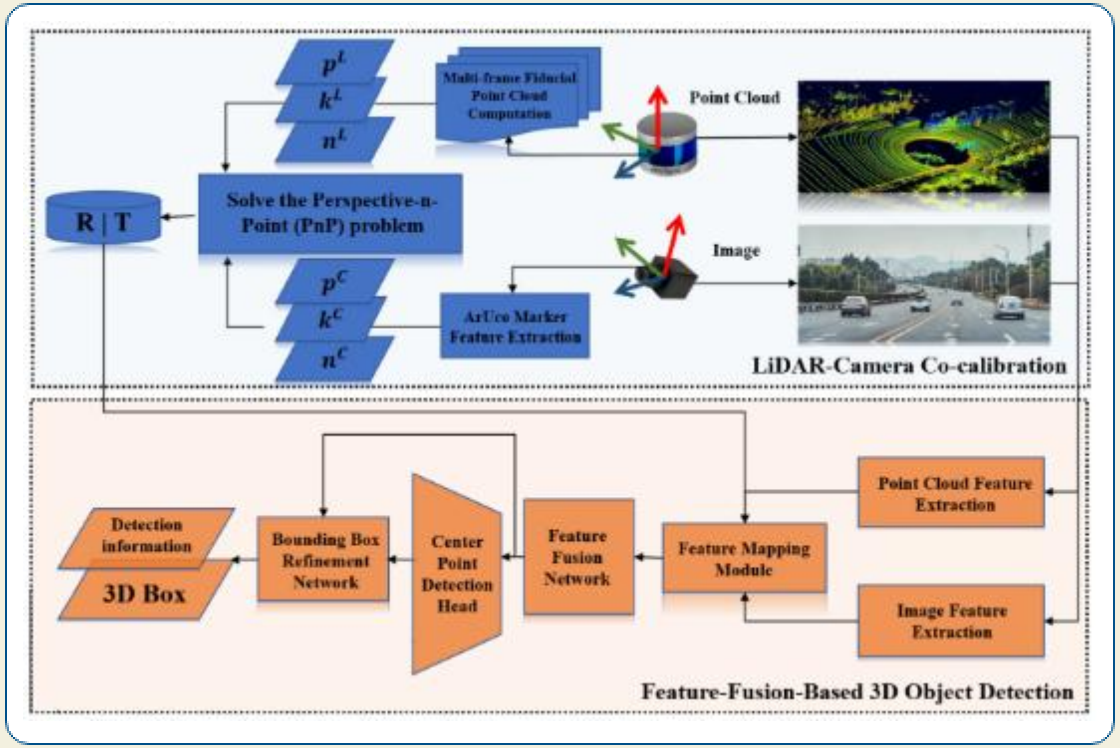
One approach is the **Factorization Method**, which can compute shape subspaces from image sequences and compare them using canonical angles.



Related Research: 3D Features

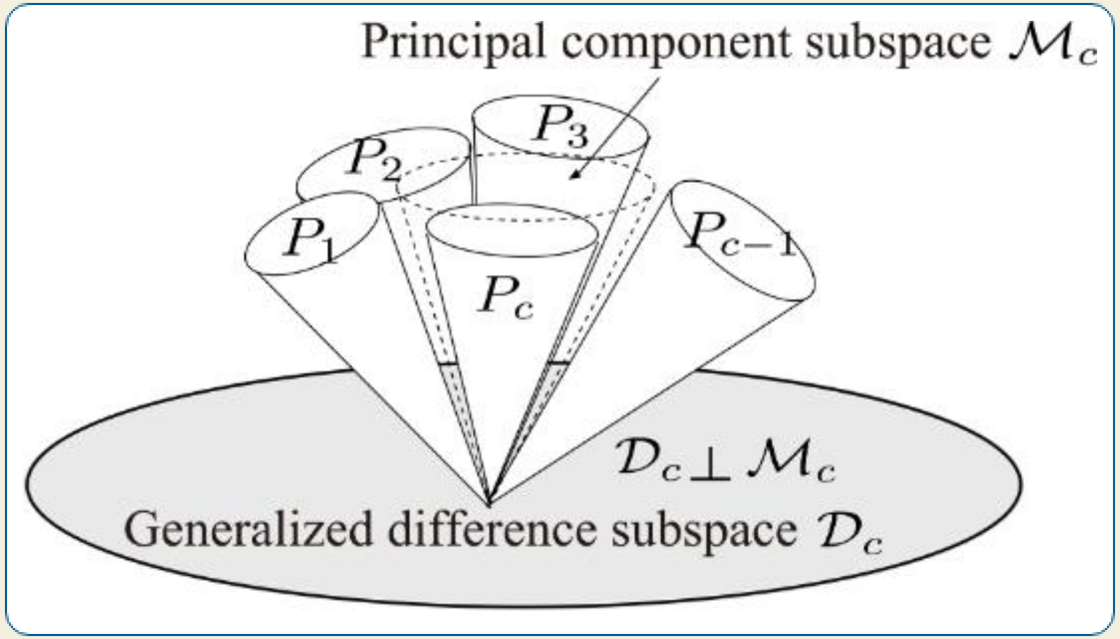
How is Shape Subspace computed?

Another approach is **Direct 3D Feature Extraction**, where shape subspaces are generated from 3D feature point sets, such as micro-features on a facial surface.



Related Research: Difference Subspace

- The **Difference Subspace (DS)** is a geometric concept that represents the components of difference between two class subspaces.
- It extends the idea of a difference vector between two vectors to the case of subspaces.
- DS is useful for analyzing and extracting local shape differences, providing a sensitive cue for recognition.



Proposed Method Overview

1. Divide Input

The input point cloud is divided into a grid of n × n patches.

2. Compute Subspace

For each patch, a centered matrix is created and SVD is applied to compute its unique Shape Subspace.

3. Compare Patches

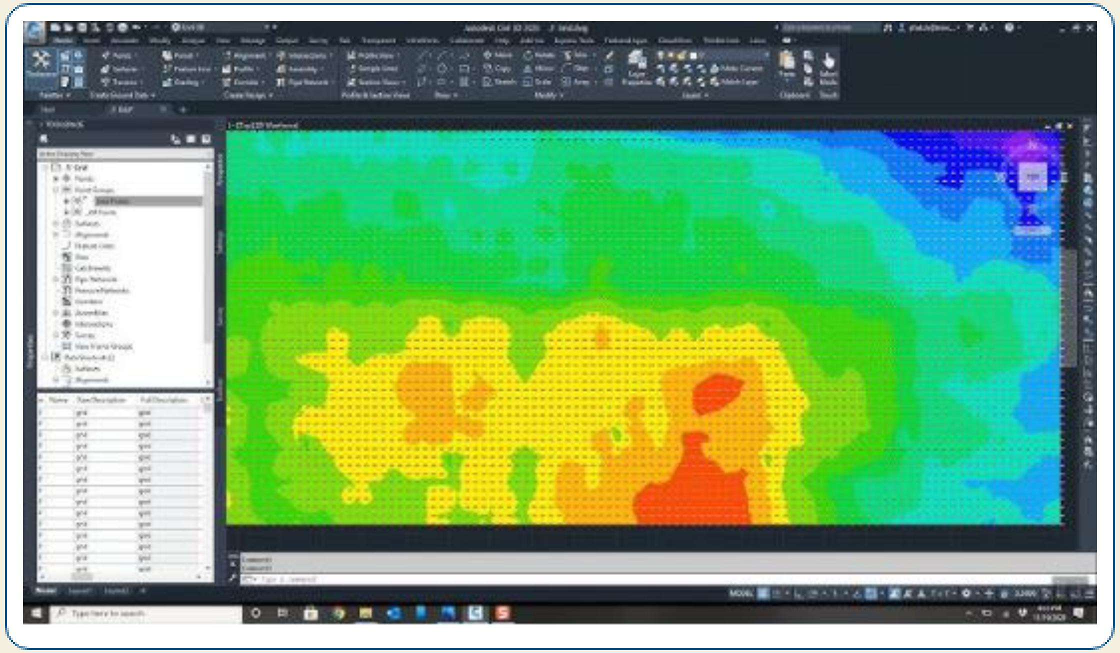
The Difference Subspace (DS) is computed between adjacent patches. A high DS magnitude indicates a local irregularity.

Experiment: Synthetic Floor

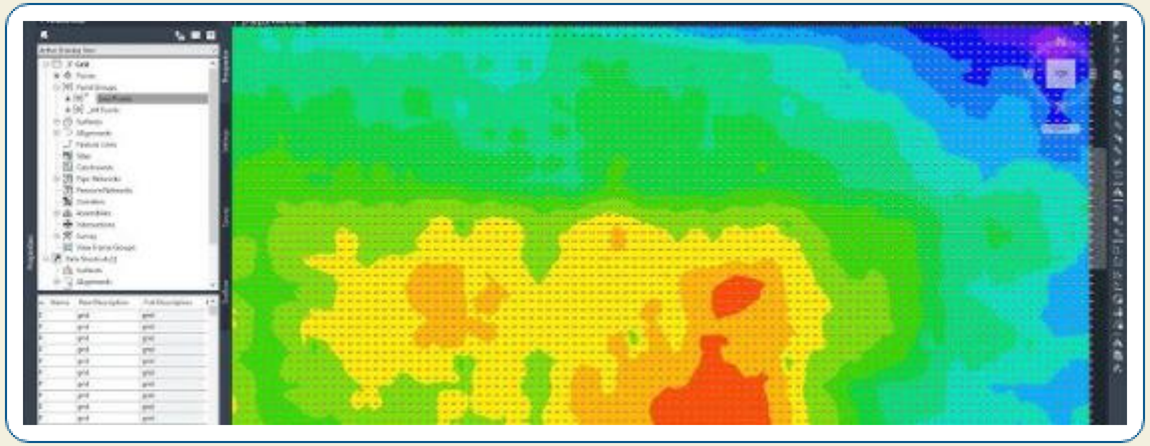
Generation of synthetic floor.

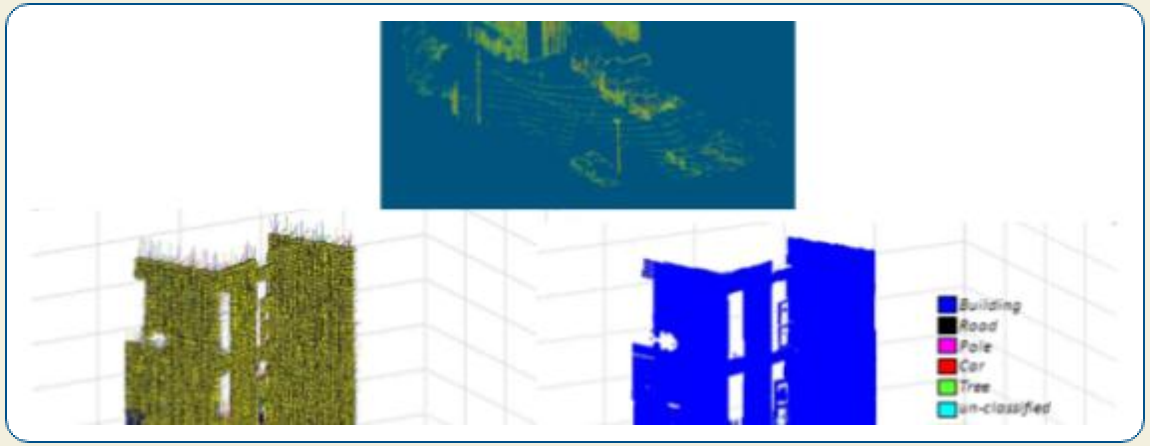
Configuration:

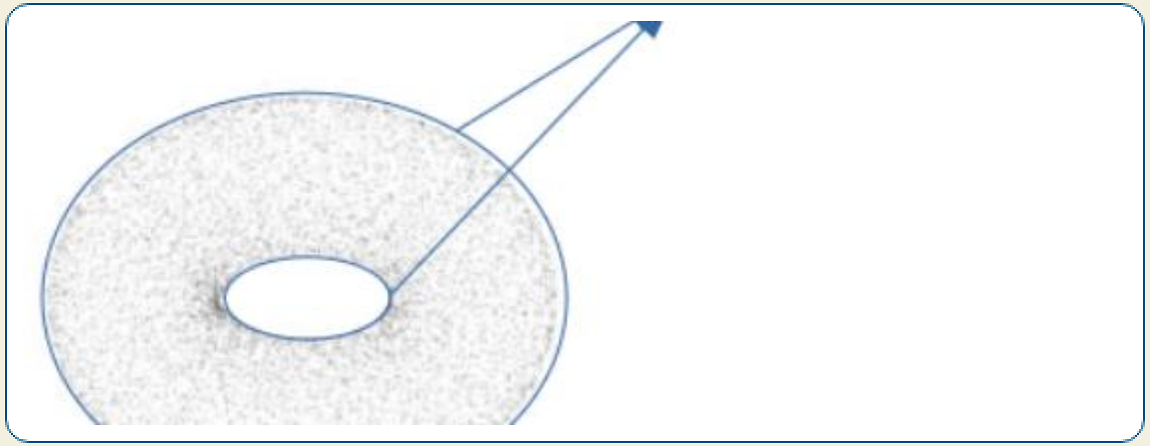
- Dimensions: 50 × 50 units
- Spacing: 1 centimeter apart
- Total points: 10,000

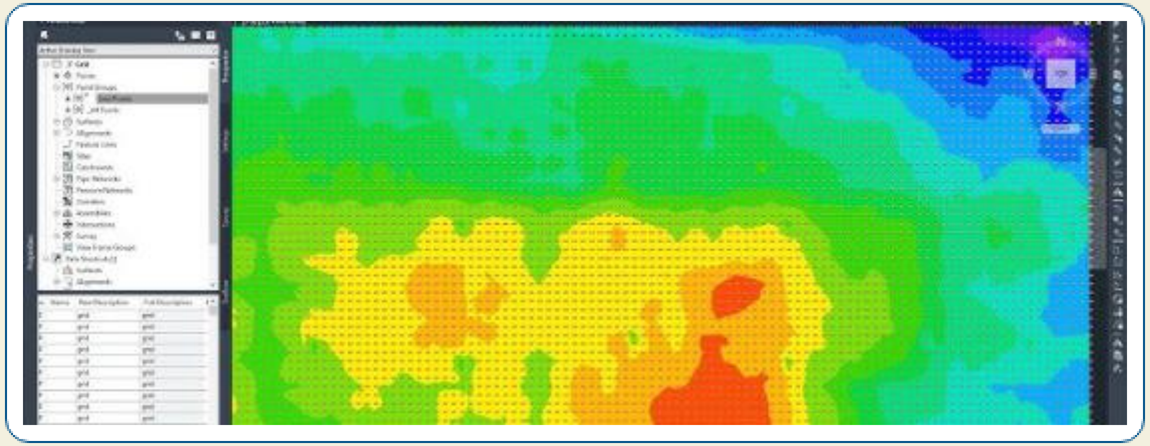


Experiment: Anomaly Patterns





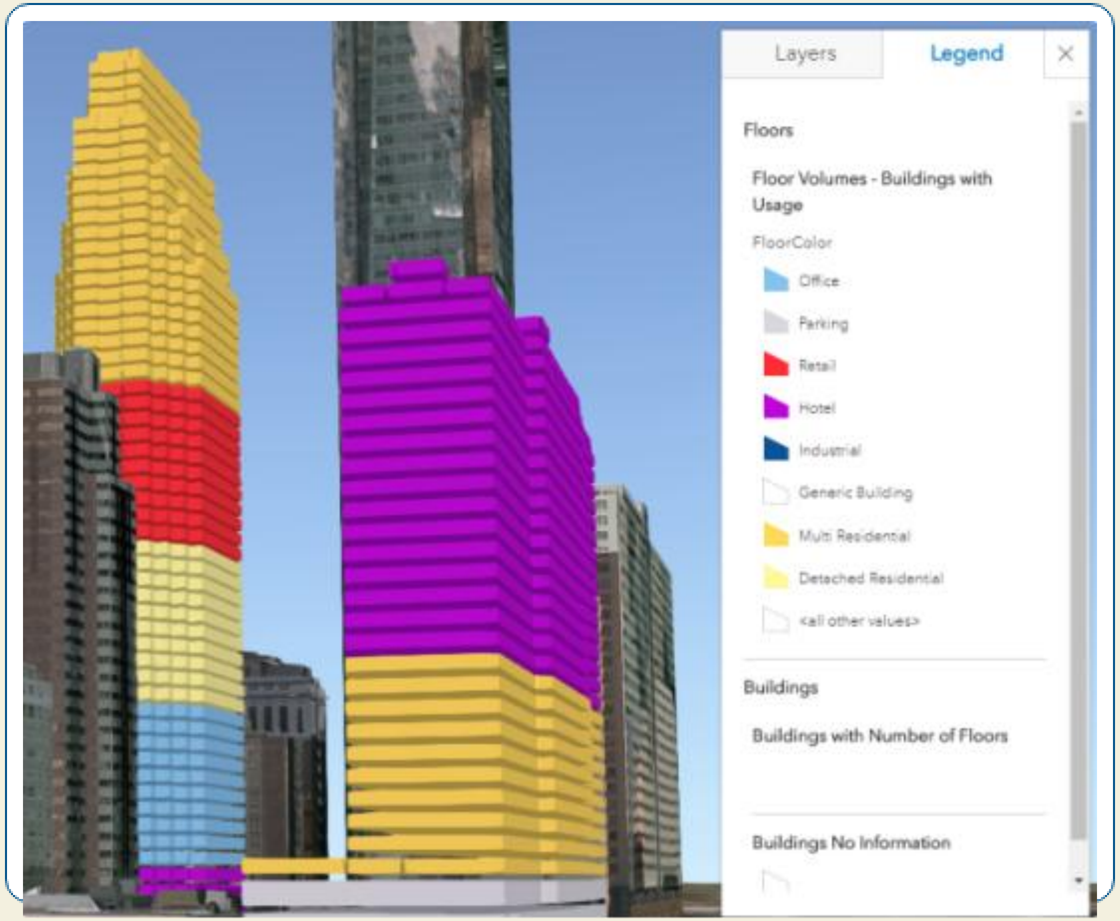




Experiment: Patch Division

Patch division process

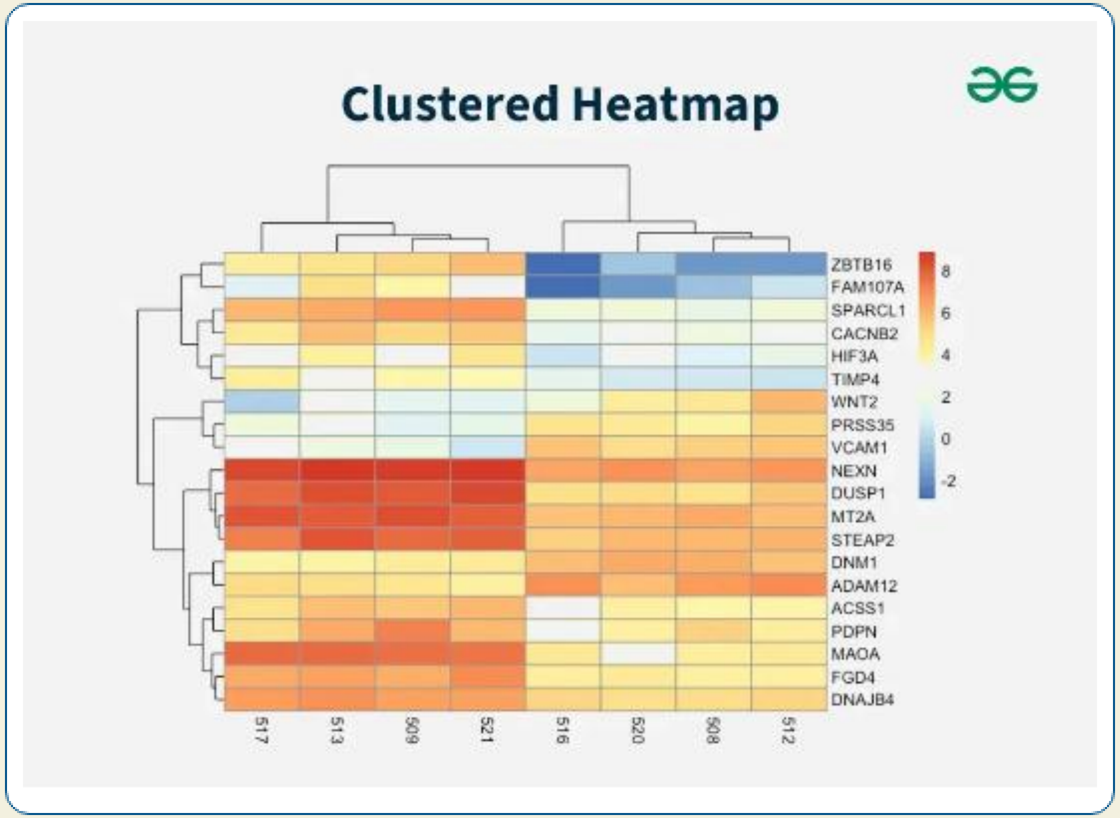
The synthetic floor is divided into a grid of patches (e.g., 5x5) for analysis. Each patch is processed independently to compute its shape subspace.



Experiment: DS Generation

DS generation for each patch

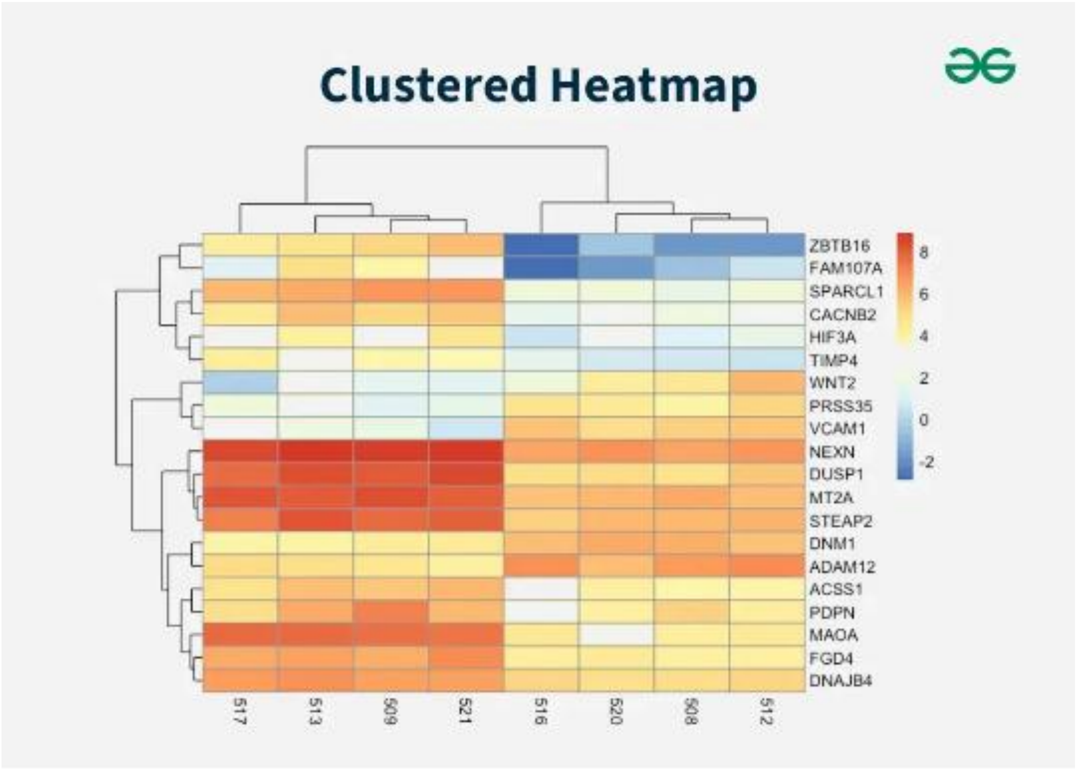
The Difference Subspace (DS) magnitude is calculated between a central patch and its neighbors (e.g., 4-neighbor or 8-neighbor approach) to quantify local shape differences.



Summary of Initial Research

Key Results

- **Pattern Detection:** Successfully detected various synthetic patterns (bumps, ridges, corners, circles, etc.)
- **Accuracy & Stability:** Smaller patches (3×3, 4×4) improved fine-detail detection.
- **Neighborhood:** Eight-neighbor comparisons enhanced stability for complex patterns.



Limitations Identified

Boundary Sensitivity

Non-overlapping patches may cut across anomalies, weakening signals near edges and leading to less accurate localization.

Limited Spatial Resolution

Sparse, discrete patches provide a coarse map, making it harder to detect and precisely locate fine-grained anomalies.

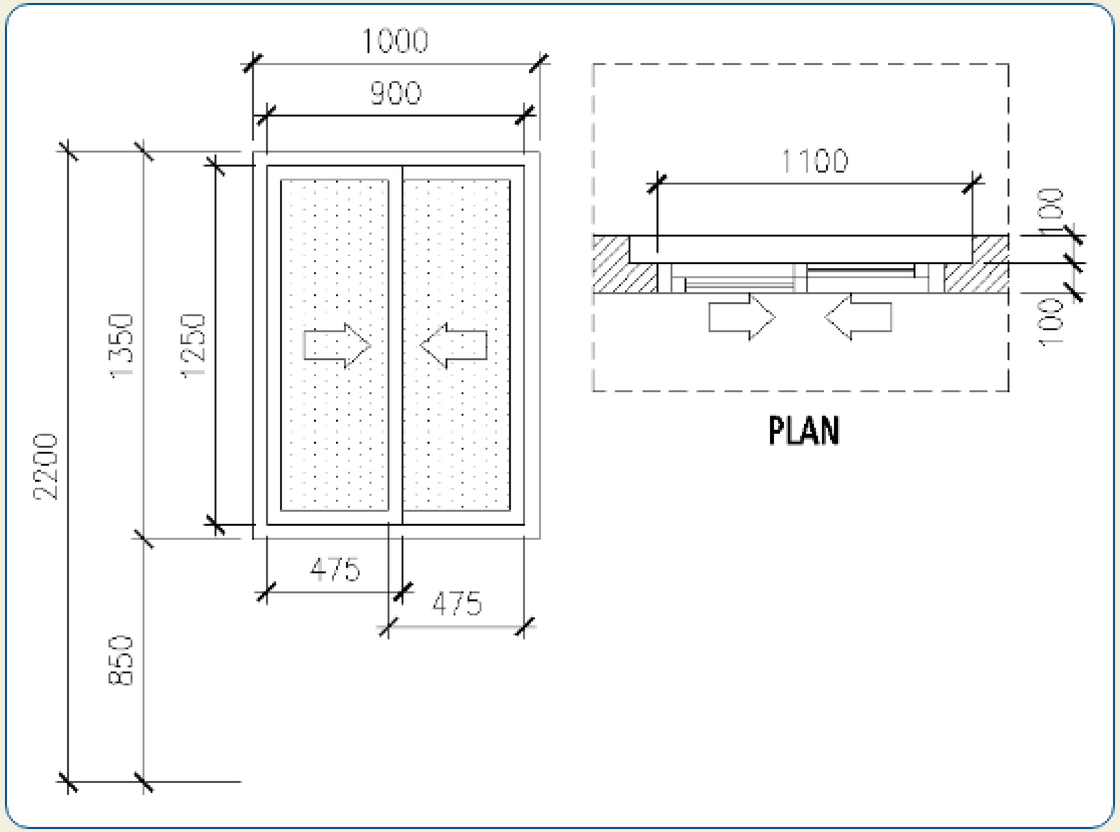
Unvalidated on Real Data

The method was only tested on clean, synthetic data. Robustness against real-world noise and irregularities was unproven.

Proposed Method: Sliding Window

Concept

- **Overlapping Regions:** Introduces analysis regions that overlap with configurable percentages (e.g., 50%).
- **Continuous Coverage:** Ensures every point is analyzed multiple times, mitigating boundary sensitivity.
- **Increased Resolution:** Generates more analysis points, achieving finer spatial resolution for anomaly localization.



Sliding Window: Methodology (1)

Given a point cloud with dimensions (W,H), a window size (w,h), and an overlap percentage α.

The stride parameters (sx, sy) — the distance the window moves — are calculated as:

$$s_x = w \cdot (1 - \alpha) \qquad s_y = h \cdot (1 - \alpha)$$

$$\frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial f_1(x)}{\partial x_1} & \frac{\partial f_1(x)}{\partial x_2} & \frac{\partial f_1(x)}{\partial x_3} & \frac{\partial f_1(x)}{\partial x_4} & \cdots & \frac{\partial f_1(x)}{\partial x_n} \\ \frac{\partial f_2(x)}{\partial x_1} & \frac{\partial f_2(x)}{\partial x_2} & \frac{\partial f_2(x)}{\partial x_3} & \frac{\partial f_2(x)}{\partial x_4} & \cdots & \frac{\partial f_2(x)}{\partial x_n} \\ \frac{\partial f_3(x)}{\partial x_1} & \frac{\partial f_3(x)}{\partial x_2} & \frac{\partial f_3(x)}{\partial x_3} & \frac{\partial f_3(x)}{\partial x_4} & \cdots & \frac{\partial f_3(x)}{\partial x_n} \\ \vdots & & & & & \\ \frac{\partial f_m(x)}{\partial x_1} & \frac{\partial f_m(x)}{\partial x_2} & \frac{\partial f_m(x)}{\partial x_3} & \frac{\partial f_m(x)}{\partial x_4} & \cdots & \frac{\partial f_m(x)}{\partial x_n} \end{bmatrix}$$

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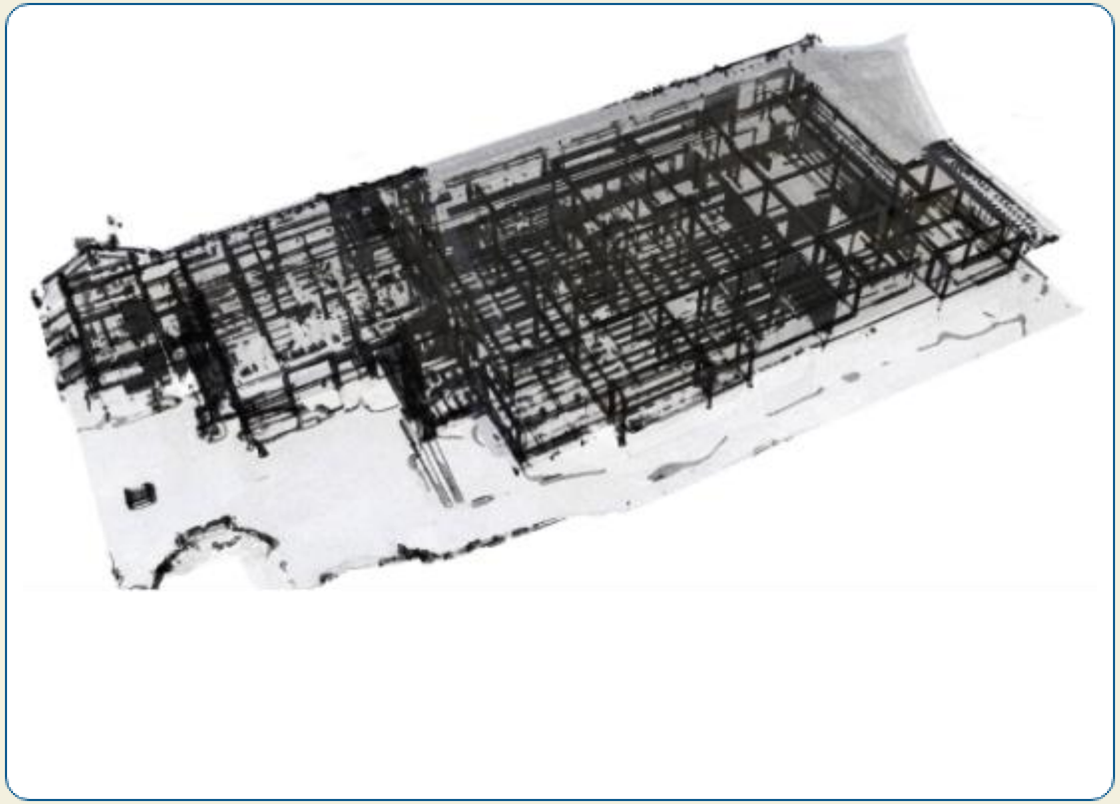
Sliding Window: Methodology (2)

The total number of windows (Nx, Ny) and the region Ri,j for each window are defined:



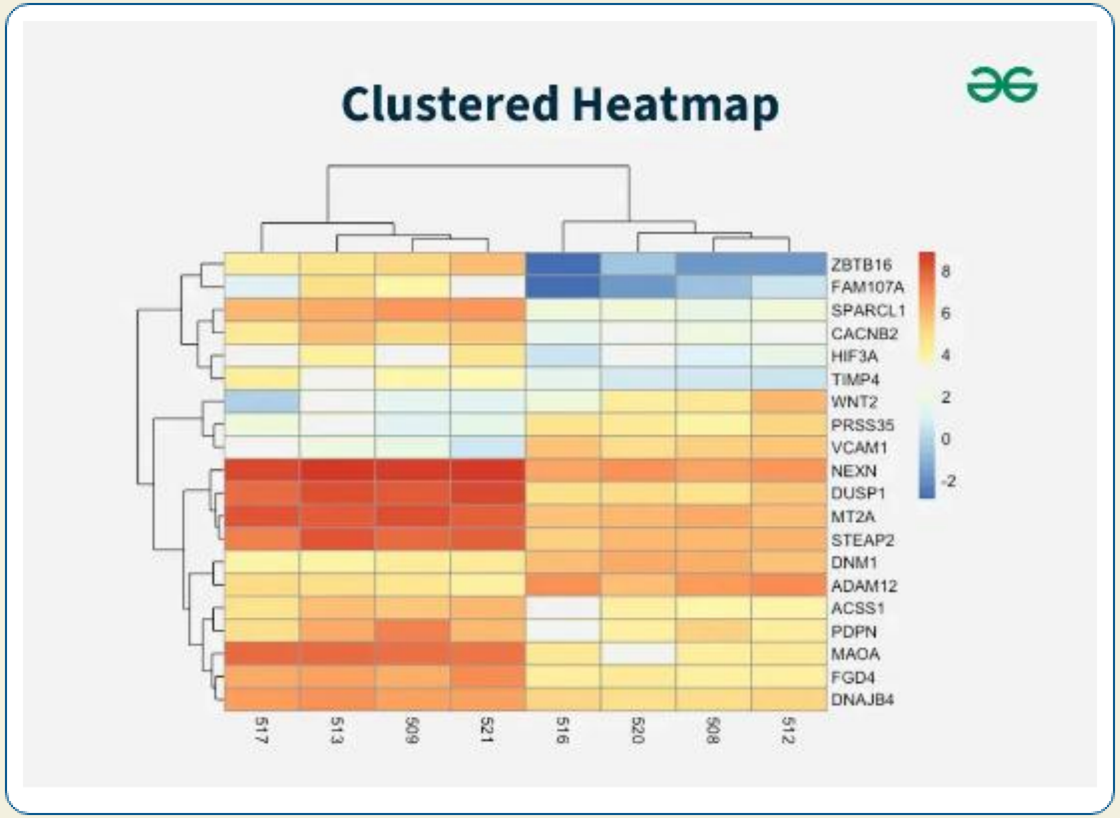
Experimental Validation: Setup

Comparison Methods

- **Fixed Patches (Baseline):** 8×8 grid (64 patches), used as reference.
- **Sliding Window (Proposed):** 50% overlap, increasing analysis density.

Data Used

- **Synthetic:** 50×50 units, ~10,000 points with predefined patterns.
- **Real-World:** PLY File, 61,914 points from a noisy floor scan.



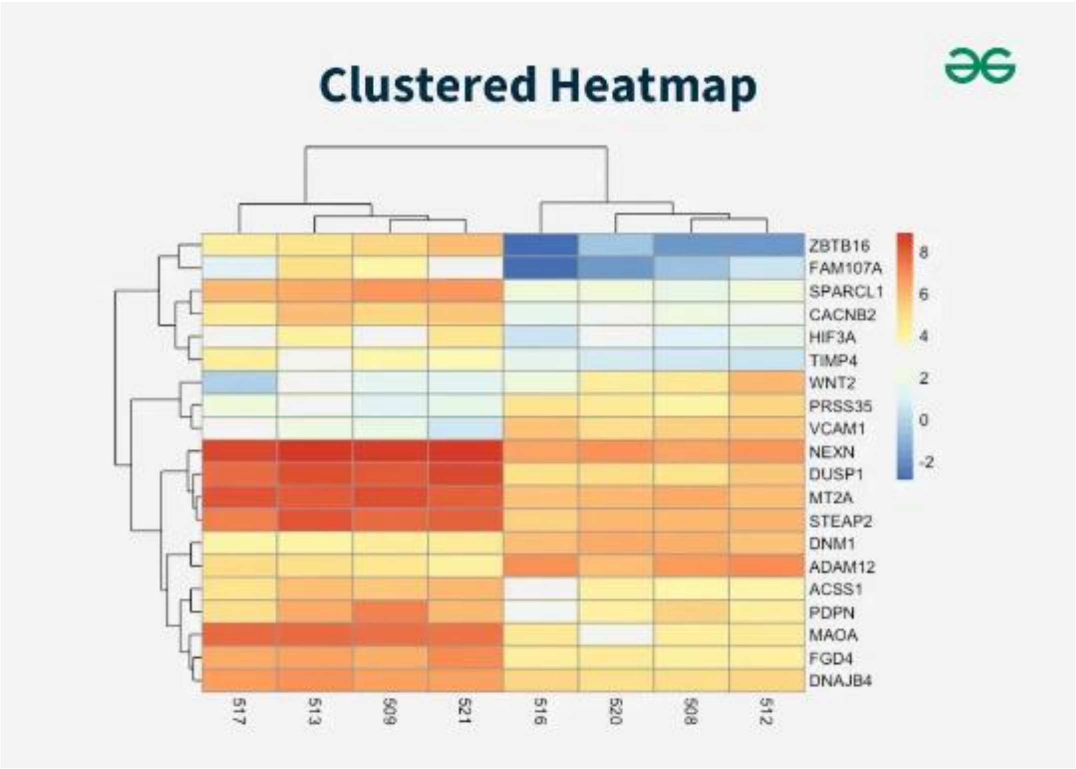
Experimental Validation: Settings

Common Settings

- **Subspace Dim:** Set to 3 for all computations.
- **Neighbors:** Used 8-neighbor DS calculation for context.
- **Sliding Overlap:** Fixed at 50% for better coverage.

Preprocessing

- **Centering:** Removed translation for geometry-only focus.
- **Min Point Filter:** Ensured enough points per window for stable SVD.



Future Work

- Adaptive Parameters
- Multi-Scale Analysis
- Performance Boost

Questions?

Thank you.

Part 2: Automatic Segmentation of Aircraft Dents

Applying Subspace Concepts to a New Domain

New Problem: Aircraft Dent Inspection

Motivation

Aircraft dent inspection is currently a manual, tedious, and subjective process, prone to human error. Dents are subtle, smooth deformations without clear boundaries.

Challenge

How do we automatically segment these dents from high-resolution 3D point clouds?

- Lack of labeled training data is a major hurdle.
- Dents have no distinctive features.
- Engineers require interpretable and trustworthy results.

Connecting Previous Work

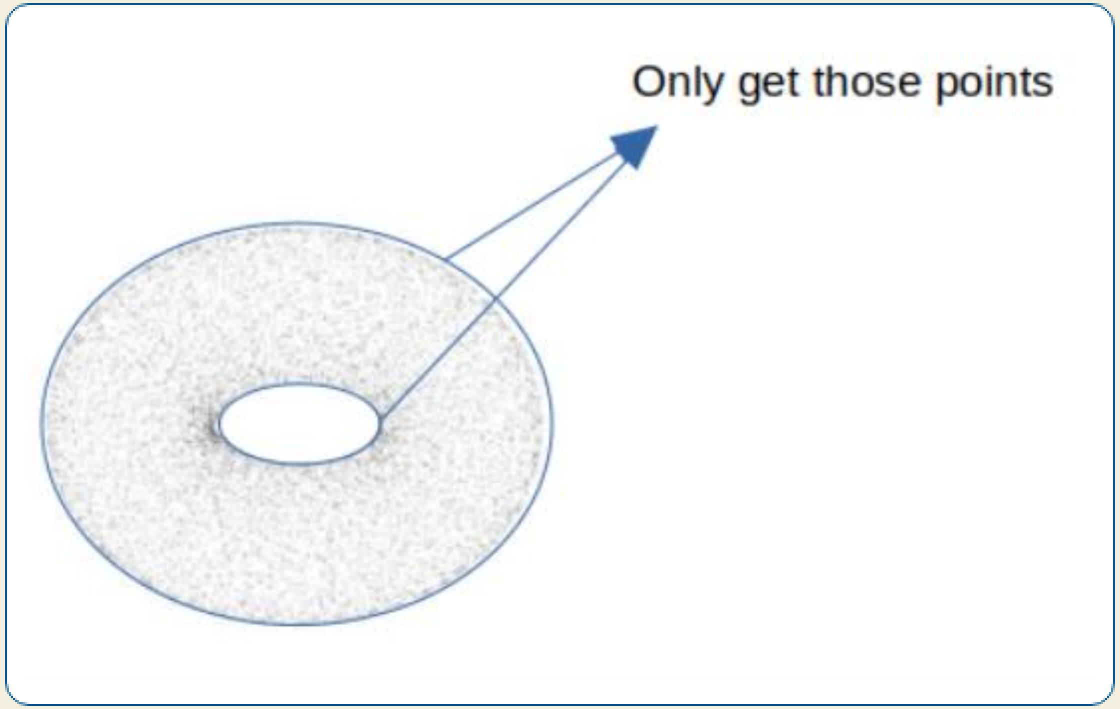
Recap of Initial Research

We explored Shape Subspace methods on synthetic floors.

- **Fixed Patch Analysis:** Successfully detected known patterns (bumps, ridges).
- **Sliding Window Extension:** Provided 3.5x higher resolution and handled noisy, real-world PLY files more effectively.

Key Finding

Method selection is data-dependent. Sliding window/patch-based methods showed more robustness to noise. This motivated exploring a similar patch-based approach for aircraft dents.



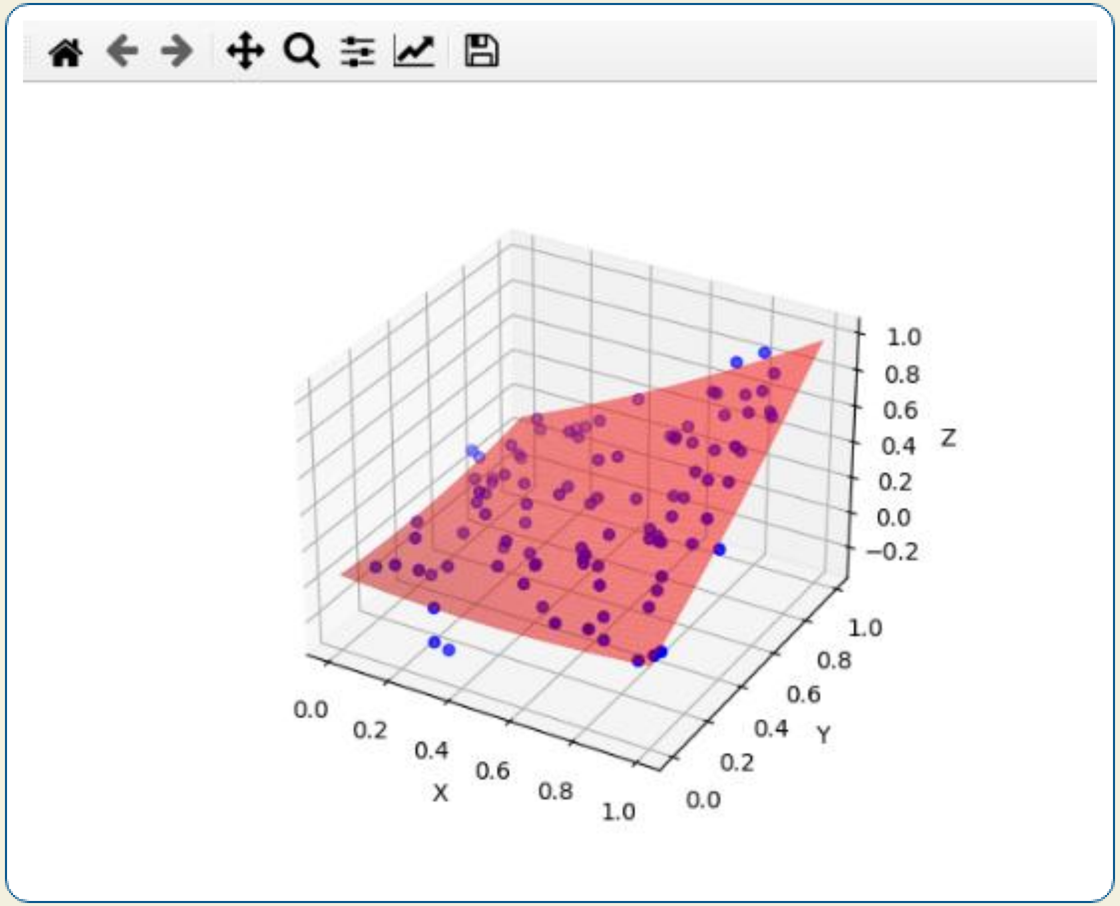
A Solution: DentNet's 2.5D Strategy

Handling 3D Data Efficiently

The DentNet paper introduces an elegant 2.5D strategy:

- **Quadric Surface Fitting:** A bivariate quadratic function, z(x,y), is fit to the 3D point cloud to capture the baseline aircraft curvature.
- **Residual Calculation:** The distance of each point from this fitted surface is calculated. Dents appear as high-residual regions.

This effectively converts the complex 3D data into a 2D residual map, which is fast, memory-efficient, and highlights deformations for a CNN.



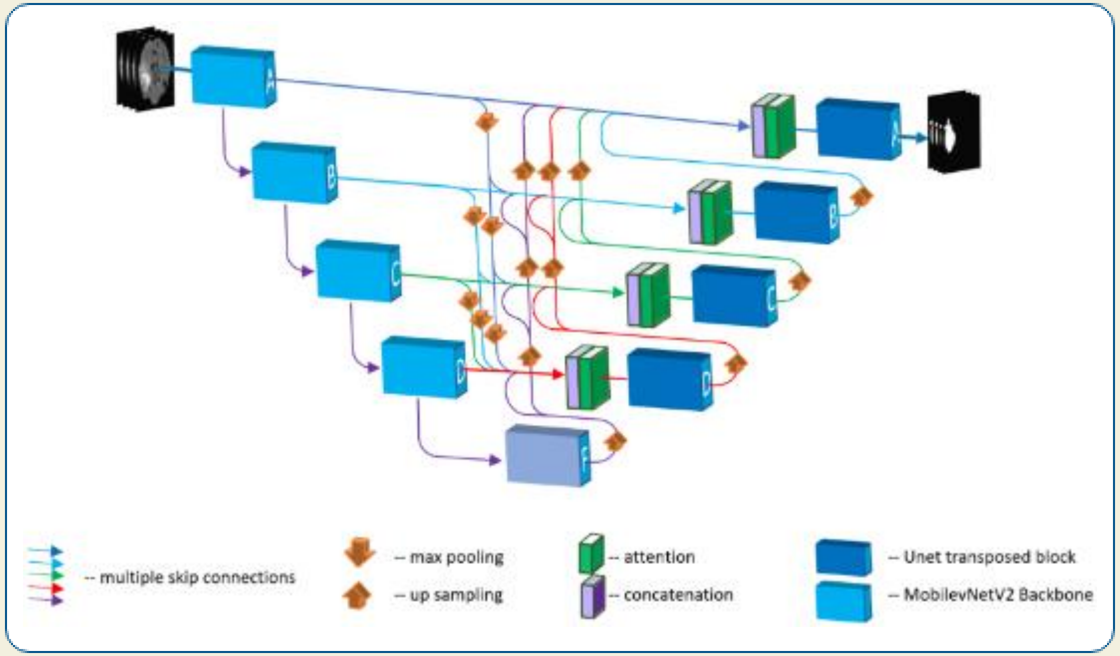
DentNet Architecture (U-Net FCN)

A U-Net for Segmentation

DentNet uses a Fully Convolutional Network based on the U-Net architecture.

- **Encoder (Downsampling):** Extracts features at multiple scales, with a 1024-channel bottleneck.
- **Decoder (Upsampling):** Reconstructs the full-size segmentation mask.
- **Skip Connections:** Propagate fine details from the encoder to the decoder, crucial for precise boundaries.

The model was trained on 30,000 synthetic samples and achieved **99.24% IoU** on synthetic validation data.



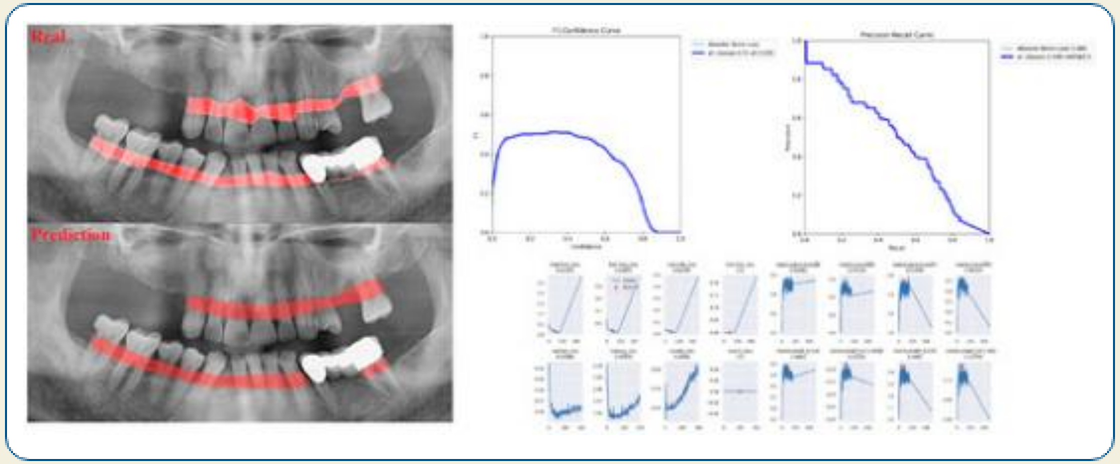
DentNet Synthetic Test Results

Performance on Synthetic Data

On a synthetic test sample, DentNet achieves **98.97% IoU**.

The output shows:

- **Input Residuals:** The dent is clearly visible as a blue depression.
- **Probability Map:** Hot colors indicate high confidence of a dent.
- **Final Prediction:** The boundaries are smooth, accurate, and (green) match the ground truth.



Our Approach: Patch-Based Subspace

1. Patch Extraction

Use a sliding window (64x64 patches, 50% overlap) to get ~700 patches per image.

2. Feature Extraction

Use pre-trained DentNet encoder (Transfer Learning) to get a 1024-D feature vector per patch.

3. Subspace Classification

Build "Dent" & "Normal" subspaces (50 PCs each) via PCA. Classify patch by distance to each subspace.

4. Reconstruction

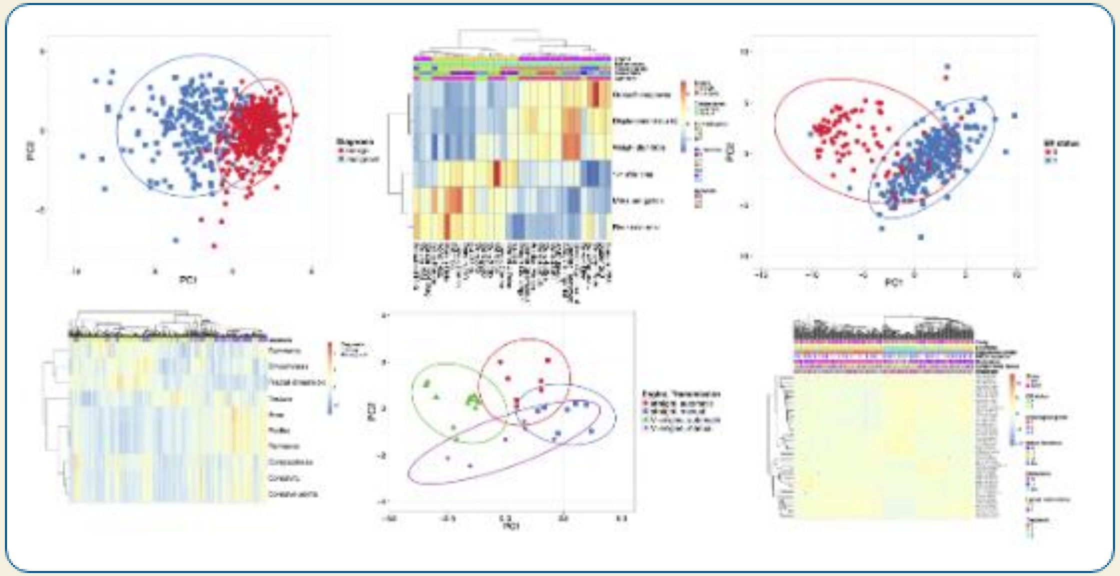
Aggregate overlapping patch predictions (voting) to generate the final segmentation mask.

Why Subspace Methods? Interpretability

Beyond the "Black Box"

Unlike a standard neural network, the subspace approach is highly interpretable.

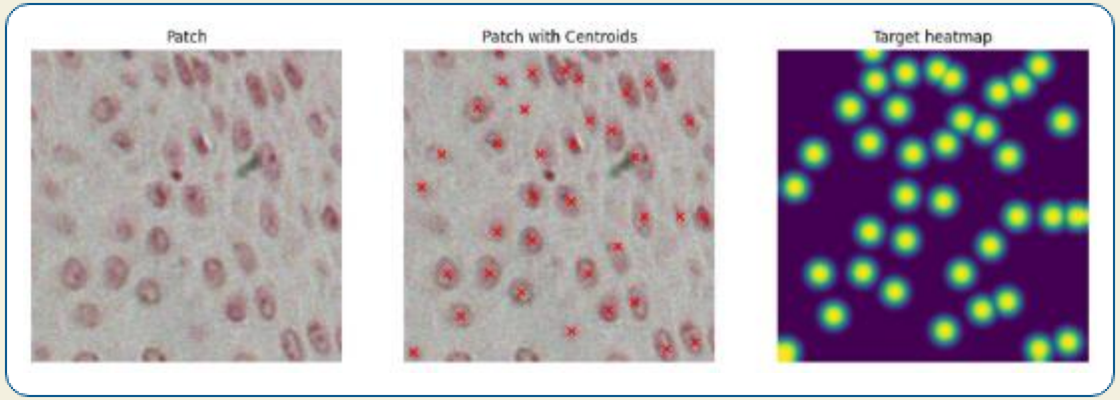
- We can visualize the learned patterns.
- We can see which features define "dent-ness".
- We can quantitatively explain *why* a patch was classified (its distance to the subspaces).
- This allows for validation with domain experts.



Comparison on Synthetic Sample

DentNet (Original)

- **98.97% IoU**
- Very accurate, smooth, circular boundaries.
- Very close to ground truth.



Patch-Based Subspace

- **74.92% IoU**
- Lower performance, blocky boundaries (due to 64x64 patches).
- Slightly overshoots the dent area.
- However, the classification is fully interpretable.

Real-World Challenges

Struggles on Real Data

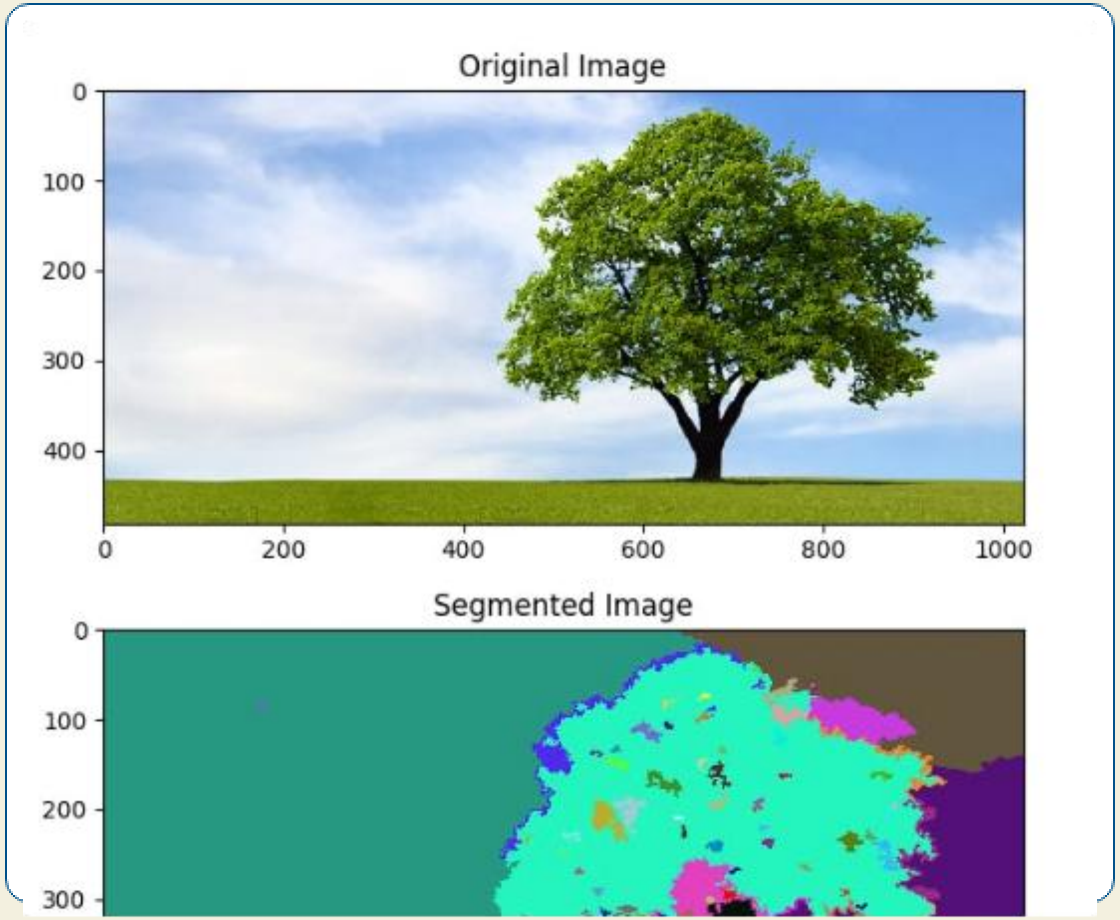
When tested on real aircraft scans, both methods struggled.

Sample A (0.78mm dent):

- **DentNet:** 41.61% False Positives. Detects scan "strip boundaries" as dents.
- **Patch-Based:** 20.65% False Positives. Better, but still problematic.

Sample C (15mm deep dent):

- Both methods show **0% detection**.



Analysis: Why Do Both Methods Fail?

Domain Gap

Training data (clean, synthetic) lacks real-world scan artifacts like stitching "strip boundaries". The models misinterpret these as anomalies.

Quadric Fitting Issue

Training: Fit *excludes* dent points (uses ground truth). **Inference:** Fit *includes* all points (no mask), "flattening" the dent and making it invisible.

Magnitude Mismatch

Training was on 1-3mm dents. Real data was 0.78mm (too shallow) or 15mm (too deep), falling outside the trained range.

Summary & Lessons Learned

- Implemented **DentNet** (99.24% synthetic IoU) and a **Patch-Based Subspace** approach.
- **Synthetic:** DentNet is highly accurate (98.97%); Patch-based is interpretable but less precise (74.92%).
- **Real Data:** Both methods struggle. The patch-based approach shows slightly better generalization to noise (20.65% FP vs 41.61% FP).
- **Connection:** This confirms our initial finding: patch-based/sliding-window methods can be more robust to noise and domain gaps.
- **Main Lesson:** The domain gap and preprocessing (quadric fitting) are the most significant challenges to solve for real-world deployment.

Future Directions

- Robust Quadric Fitting
- Augment Training Data
- Domain Adaptation

Questions?

Thank you.

Image Sources

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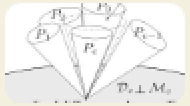

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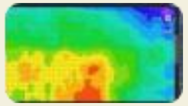

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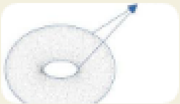

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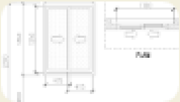

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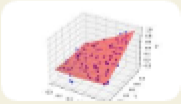

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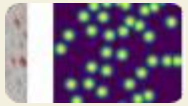

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